# ZDC autoresearch, Phase 5: $\Lambda \to n\pi^0$ on MSF events

Find more $\Lambda \to n\pi^0$ in EICrecon on meson-structure events without a dirtier $\Lambda$ sample

Started 2026-11-08 · one slide per try · rebuilt from the run's result files

# Phase 5 tries — Λ found on the search halves, step by step

| try | change | 9x275 | 9x130 | score | purity | z | verdict |
|---|---|---|---|---|---|---|---|
| Y1 | B0 ECal clusters split at local maxima | 943 → 944 | 66 → 71 | 1009 → 1015 | 19.8 → 19.0 % | +1.2 | INCONCLUSIVE |
| Y2 | the Λ vertex with each daughter's lever arm | 943 → 953 | 66 → 83 | 1009 → 1036 | 19.8 → 20.4 % | +4.2 | ACCEPTED |
| Y4 | B0 splitting retried on top of Y2 | 953 → 975 | 83 → 123 | 1036 → 1098 | 20.4 → 20.0 % | +7.0 | ACCEPTED |
| Y5 | the ZDC pairing with a photon in B0 | 975 → 1164 | 123 → 147 | 1098 → 1311 | 20.0 → 23.8 % | +13.3 | ACCEPTED |
| Y6 | sharper energy sharing between split B0 clusters | 1164 → 1209 | 147 → 182 | 1311 → 1391 | 23.8 → 24.7 % | +8.1 | ACCEPTED |
| Y7 | the B0 photon energy law fitted on a B0 photon gun | 1209 → 1238 | 182 → 190 | 1391 → 1428 | 24.7 → 25.1 % | +3.7 | ACCEPTED |
| Y8 | B0 split sharing narrowed again, 0.5 → 0.3 crystal | 1238 → 1252 | 190 → 194 | 1428 → 1446 | 25.1 → 25.3 % | +3.4 | ACCEPTED |
| Y9 | the ZDC pairing with both photons in B0 | 1252 → 1267 | 194 → 219 | 1446 → 1486 | 25.3 → 26.7 % | +5.2 | ACCEPTED |
| Y10 | the Λ-mass term of the final choice, 112 → 40 MeV | 1267 → 1267 | 219 → 219 | 1486 → 1486 | 26.7 → 26.7 % | +0.0 | REJECTED (no effect) |

**Score 1009 → 1486 (+47 %), purity 19.8 → 26.7 %; each row is paired against the best of its time.**

# Phase 5 tries in Phase 4's columns — S4a_tm per thrown $\Lambda \to n\pi^0$ on the D4 gun

| S4a_tm per thrown $\Lambda \to n\pi^0$ | mean 100–275 GeV | at 275 GeV | MSF found, 9x275 / 9x130 | verdict |
|---|---|---|---|---|
| R1 (PR 3064), start of Phase 4 | 0.404 % | 0.832 % | — | |
| X9, end of Phase 4 = start of Phase 5 | 7.436 % | 14.101 % | 943 / 66 | |
| Y1: B0 ECal clusters split at local maxima | 7.436 % | 14.101 % | 944 / 71 | INCONCLUSIVE |
| Y2: the Λ vertex with each daughter's lever arm | 7.439 % | 14.117 % | 953 / 83 | ACCEPTED |
| Y4: B0 splitting retried on top of Y2 | 7.439 % | 14.117 % | 975 / 123 | ACCEPTED |
| Y5: the ZDC pairing with a photon in B0 | 7.439 % | 14.117 % | 1164 / 147 | ACCEPTED |
| Y6: sharper energy sharing between split B0 clusters | 7.439 % | 14.117 % | 1209 / 182 | ACCEPTED |
| Y7: the B0 photon energy law fitted on a B0 photon gun | 7.439 % | 14.117 % | 1238 / 190 | ACCEPTED |
| Y8: B0 split sharing narrowed again, 0.5 → 0.3 crystal | 7.439 % | 14.117 % | 1252 / 194 | ACCEPTED |
| Y9: the ZDC pairing with both photons in B0 | 7.439 % | 14.117 % | 1267 / 219 | ACCEPTED |
| Y10: the Λ-mass term of the final choice, 112 → 40 MeV | 7.439 % | 14.117 % | 1267 / 219 | REJECTED (no effect) |
| ceiling (perfect separation, P1b resolution) | 14.1 % | 27.8 % | reachable 43.6 % / 8.0 % of expected | |

# Phase 5 score — yield and purity per beam energy

```
N_expected = 0.358 × generated Λ   (one Λ per event)
N_found    = events whose true Λ → n pi0 is rebuilt:
             candidate within ±30 MeV of m_Λ, and
             n, γ, γ clusters ≥ 80 % from the true
             neutron and the two true photons
N_window   = events with any candidate in the window

yield  = N_found / N_expected
purity = N_found / N_window
```

| X9 on D6c (40 000 events per beam) | 9x275 | 9x130 |
|---|---|---|
| N_found / N_expected | 1840 / 14320 | 129 / 14320 |
| yield | 12.85 ± 0.28 % | 0.90 ± 0.08 % |
| reachable (n in ZDC, both γ read) | 43.6 % | 8.0 % |
| N_found / N_window | 1840 / 8749 | 129 / 1408 |
| purity | 21.03 ± 0.44 % | 9.16 ± 0.77 % |

**Selector (463): keep a change when**

- the score N_found(9x275) + N_found(9x130) on the search halves D6s rises: on identical Λ, b found only by the current best, c only by the change, $z = (c - b) / \sqrt{(b + c)} \geq 3$
- and the pooled purity does not fall by more than 2σ: $z_P = (P_{new} - P_{best}) / \sqrt{(\sigma_{new}^2 + \sigma_{best}^2)} \geq -2$
- D6h (files 41–80) is held out: every 5 tries the current best must reproduce its gain there at > 2σ
- fair play F1–F3 of Phase 4 (460) on the MSF outputs; a FAIL voids the try

**Yield is pushed up; purity is a floor. A change that only cleans the sample is reported, not kept.**

# Where $\Lambda \rightarrow n\pi^0$ land and what X9 finds in each regime

| regime (share of N_expected; X9 found in it) | 9x275 | 9x130 |
|---|---|---|
| all three in the ZDC | 27.9 %; 1472 found (36.8 %) | 1.3 %; 62 found (34.3 %) |
| n ZDC, γ B0 + γ ZDC | 11.0 %; 366 found (23.3 %) | 2.1 %; 66 found (22.0 %) |
| n ZDC, γ γ in B0 | 4.0 %; 1 found (0.2 %) | 3.5 %; 1 found (0.2 %) |
| n ZDC, a γ in EEMC-P / LFHCAL | 0.7 %; 0 found (0.0 %) | 1.2 %; 0 found (0.0 %) |
| n ZDC, a γ lost | 25.6 %; 0 found (0.0 %) | 10.1 %; 0 found (0.0 %) |
| n in B0 | 6.6 %; 1 found (0.1 %) | 30.0 %; 0 found (0.0 %) |
| n lost or elsewhere | 1.1 %; 0 found (0.0 %) | 23.3 %; 0 found (0.0 %) |
| Λ interacts in material / does not decay | 17.2 % / 6.4 % | 27.2 % / 0.2 % |

**At 9x275 the software can still win in the first four rows; at 9x130 the neutron misses the ZDC for 53 %.**

# Y1 — B0 ECal clusters split at local maxima

### Change

- B0 island clustering keeps one cluster per island today (splitCluster false)
- with both photons in B0 (4.0 % of expected at 9x275) the pair is merged for 93 % of the Λ
- split at every hit above 30 MeV that exceeds its 8 neighbours
- shared hits weighted by exp(−d / 1 crystal) × peak energy (stock: η-φ distance, scale 1)
- three parameters on the X9 install; no code change

### Prediction (written before launch)

- N_found on the search halves: 9x275 ≈ 920 → 960, 9x130 ≈ 65 → 95; purity flat or down < 1 point

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 943 | 944 |
| 9x275 purity | 21.5 % | 20.6 % |
| 9x130 N_found | 66 | 71 |
| 9x130 purity | 9.4 % | 9.5 % |

**paired: b 9, c 15, z +1.22; purity z_P -1.01**

INCONCLUSIVE

fair play PASS

# Y2 — the Λ vertex with each daughter's lever arm

### Change

- the Λ line through the IP had slope $\Sigma\, p_i x_i \,/\, \Sigma\, p_i z_i$: right when n, γ, γ hit one depth
- a B0 photon ($z \approx 7$ m) beside ZDC daughters ($z \approx 35$ m) was weighted down by ~7/35
- now, for a vertex at depth z: $s_\perp = \Sigma\, w_i x_{i\perp} \,/\, (P + z\, \Sigma\, w_i)$, $w_i = p_i \,/\, (z_i - z)$, at every bisection step
- the depth search ends short of the nearest impact point (it ran on to the ZDC)
- 33 lines in LambdaReconstruction.cc; all-ZDC events keep the stock result

### Prediction (written before launch)

- 9x275 920 → ~990, 9x130 ~65 → ~80; all-ZDC unchanged within ±10; purity up slightly

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 943 | 953 |
| 9x275 purity | 21.5 % | 21.7 % |
| 9x130 N_found | 66 | 83 |
| 9x130 purity | 9.4 % | 11.9 % |

**paired: b 7, c 34, z +4.22; purity z_P +0.71**

**ACCEPTED** | **fair play PASS**

# Y4 — B0 splitting retried on top of Y2

### Change

- Y1 resolved both B0 photons for ~50 more Λ (9x275) and ~65 (9x130), but their mass failed
- Y2 made the γγ-in-B0 regime count: 0 → 5 found at 9x275, 1 → 15 at 9x130
- Y1's three B0 island parameters on the Y2 install

### Prediction (written before launch)

- γγ in B0 found 5 → ~20 and 15 → ~35; N_found 953 → ~968, 83 → ~103; purity flat

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 953 | 975 |
| 9x275 purity | 21.7 % | 20.8 % |
| 9x130 N_found | 83 | 123 |
| 9x130 purity | 11.9 % | 15.3 % |

**paired: b 8, c 70, z +7.02; purity z_P -0.50**

**ACCEPTED** **fair play PASS**

# Y5 — the ZDC pairing with a photon in B0

### Change

- the ZDC pairing assumed both photons in the ZDC: with one photon in B0 it still wrote two ZDC photons
- now each tagged ZDC cluster with each B0 photon candidate is a hypothesis too (Y2's vertex)
- a winning mixed hypothesis writes one ZDC photon; the rest of the ZDC is the neutron
- LambdaReconstruction pairs that photon with the B0 photon
- switch lambdaPairingMixed, off by default; off reproduces Y4 exactly

### Prediction (written before launch)

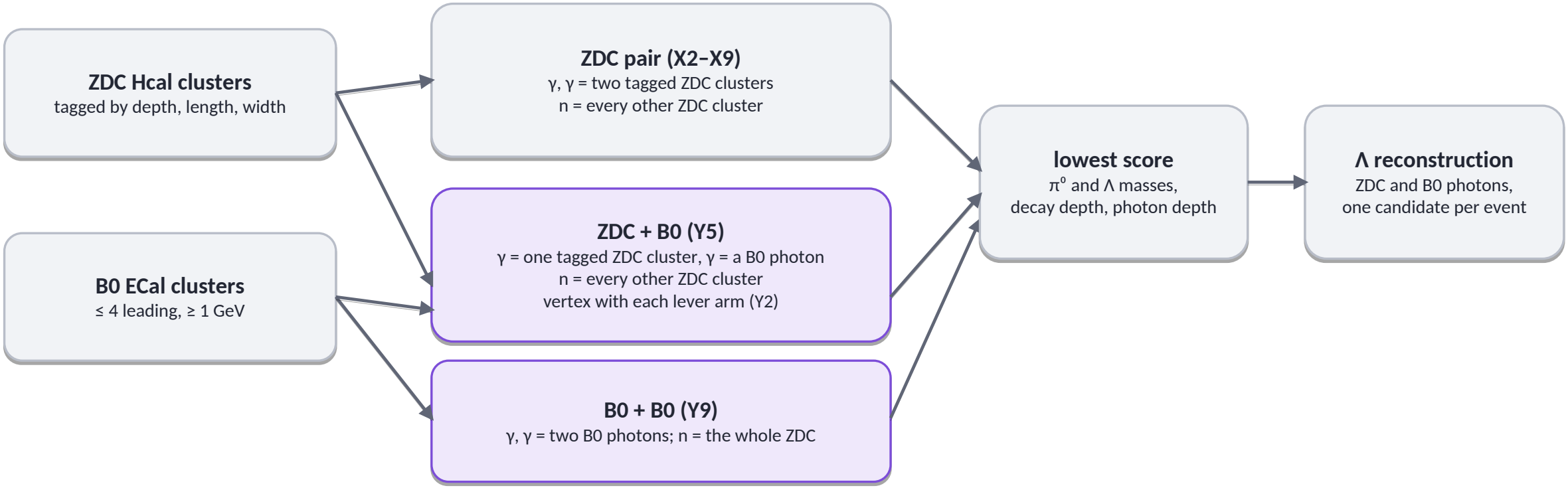
- emulated mixed regime 195 → 383 (9x275), 32 → 55 (9x130); N_found 975 → ~1150, 123 → ~145; purity up

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 975 | 1164 |
| 9x275 purity | 20.8 % | 24.6 % |
| 9x130 N_found | 123 | 147 |
| 9x130 purity | 15.3 % | 19.1 % |

**paired: b 22, c 235, z +13.29; purity z_P +4.84**

**ACCEPTED** **fair play PASS**

# Y5 and Y9 — photon hypotheses in the ZDC pairing




**Why**

- with one photon in B0 the old pairing still wrote two ZDC photons: the true one with a neutron fragment, or two fragments
- their energy left the neutron, and the true triplet could not form: another triplet chosen (186) or the ZDC photon missing (144) of 803
- emulation: Y5 383 instead of 195 in γ B0 + γ ZDC (474); Y9 +49 net with both photons in B0 (475)

# Y6 — sharper energy sharing between split B0 clusters

### Change

- with both photons in B0, Y5 still misses a photon for 198 of 264 Λ (9x275) and 151 of 264 (9x130)
- the photon's best B0 cluster is shared below purity 0.8 (~50 %) or merged with the other photon (~45 %)
- split hits are shared by exp(−d / s) × peak energy; s = 1.0 → 0.5 crystal
- one parameter on the Y5 install

### Prediction (written before launch)

- γγ-in-B0 found 33 → ~50 (9x275), 57 → ~80 (9x130); N_found 1164 → ~1180, 147 → ~170; purity flat

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 1164 | 1209 |
| 9x275 purity | 24.6 % | 25.0 % |
| 9x130 N_found | 147 | 182 |
| 9x130 purity | 19.1 % | 23.5 % |

**paired: b 9, c 89, z +8.08; purity z_P +1.17**

**ACCEPTED** **fair play PASS**

# Y7 — the B0 photon energy law fitted on a B0 photon gun

### Change

- stock law 0.99·E^1.14 reads gun photons 11–48 % high at 5–40 GeV
- refit on D2b_photon_B0 (2–40 GeV, 7–20 mrad) with Y6's B0 clustering: 1.0888·E^1.0058
- every gun energy within 0.2 % of true; one parameter on Y6

### Prediction (written before launch)

- γγ in B0 up, γ B0 + γ ZDC down (the stock overestimate cancelled the ZDC photon's 0.78); net small, sign uncertain

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 1209 | 1238 |
| 9x275 purity | 25.0 % | 25.2 % |
| 9x130 N_found | 182 | 190 |
| 9x130 purity | 23.5 % | 23.9 % |

**paired: b 31, c 68, z +3.72; purity z_P +0.38**

**ACCEPTED** **fair play PASS**

# Y8 — B0 split sharing narrowed again, 0.5 → 0.3 crystal

### Change

- Y6 (1.0 → 0.5 crystal) gained +80, z +8.1
- both photons in B0 is still the regime with the most B0 losses: 87 and 97 of 264 found
- one parameter on Y7

### Prediction (written before launch)

- N_found 1238 → ~1250, 190 → ~200; the gain may stop or turn (tail energy to the wrong maximum)

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 1238 | 1252 |
| 9x275 purity | 25.2 % | 25.4 % |
| 9x130 N_found | 190 | 194 |
| 9x130 purity | 23.9 % | 24.3 % |

**paired: b 5, c 23, z +3.40; purity z_P +0.28**

**ACCEPTED** **fair play PASS**

# Y9 — the ZDC pairing with both photons in B0

### Change

- with both photons in B0 another triplet wins for 46 (9x275) and 53 (9x130) of 264 Λ
- Y5's pairing still writes ZDC fragments as photons there and takes their energy from the neutron
- third hypothesis: every pair of B0 photon candidates, the whole ZDC as the neutron
- if it wins, no ZDC photon is written; switch lambdaPairingExternalPair, off reproduces Y8

### Prediction (written before launch)

- emulated net +21 / +28 in γγ-in-B0, −4 / −3 in γ B0 + γ ZDC; N_found 1252 → ~1268, 194 → ~219

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 1252 | 1267 |
| 9x275 purity | 25.4 % | 26.4 % |
| 9x130 N_found | 194 | 219 |
| 9x130 purity | 24.3 % | 28.7 % |

**paired: b 10, c 50, z +5.16; purity z_P +1.76**

**ACCEPTED** **fair play PASS**

# Y10 — the Λ-mass term of the final choice, 112 → 40 MeV

**Change**

- LambdaReconstruction ranks triplets across ZDC, B0 and EEMC-P by $\pi^0$ and Λ mass $\chi^2$
- its Λ term had the width 0.1 $m_\Lambda$ = 112 MeV; now 40 MeV, as the ZDC pairing
- after Y5 and Y9 another triplet wins for only 22 Λ on both search halves

**Prediction (written before launch)**

- N_found 1267 → ~1272, 219 → ~222; N_window up, purity down < 1 point

| search half | best | this try |
|---|---|---|
| 9x275 N_found | 1267 | 1267 |
| 9x275 purity | 26.4 % | 26.4 % |
| 9x130 N_found | 219 | 219 |
| 9x130 purity | 28.7 % | 28.6 % |

**paired: b 0, c 0, z +0.00; purity z_P -0.02**

REJECTED

fair play PASS

# Checkpoint 1 — held-out MSF files and the D4 gun, X9 → Y6

| N_found (yield) | X9, search | Y6, search | X9, held out | Y6, held out | reachable |
|---|---|---|---|---|---|
| 9x275 | 943 (13.17 %) | 1209 (16.89 %) | 897 (12.53 %) | 1151 (16.08 %) | 43.6 % |
| 9x130 | 66 (0.92 %) | 182 (2.54 %) | 63 (0.88 %) | 165 (2.30 %) | 8.0 % |
| purity, held out (pooled) | | | 19.0 % | 23.8 % | |

**held out: REPRODUCES**

**b 30, c 386, z +17.45; purity z_P +6.06**

**D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on)**

- X9 (end of Phase 4): 7.44 % mean, 14.10 % at 275 GeV;  Y6: 7.44 % mean, 14.12 % at 275 GeV (ceilings 14.1 % / 27.8 %)

**The search halves gained 38 %, the held-out halves 37 %: the accepted steps are no overfit.**

# Checkpoint 2 — held-out MSF files and the D4 gun, Y6 → Y9

| N_found (yield) | Y6, search | Y9, search | Y6, held out | Y9, held out | reachable |
|---|---|---|---|---|---|
| 9x275 | 1209 (16.89 %) | 1267 (17.70 %) | 1151 (16.08 %) | 1206 (16.84 %) | 43.6 % |
| 9x130 | 182 (2.54 %) | 219 (3.06 %) | 165 (2.30 %) | 205 (2.86 %) | 8.0 % |
| purity, held out (pooled) | | | 23.8 % | 25.8 % | |

**held out: REPRODUCES**

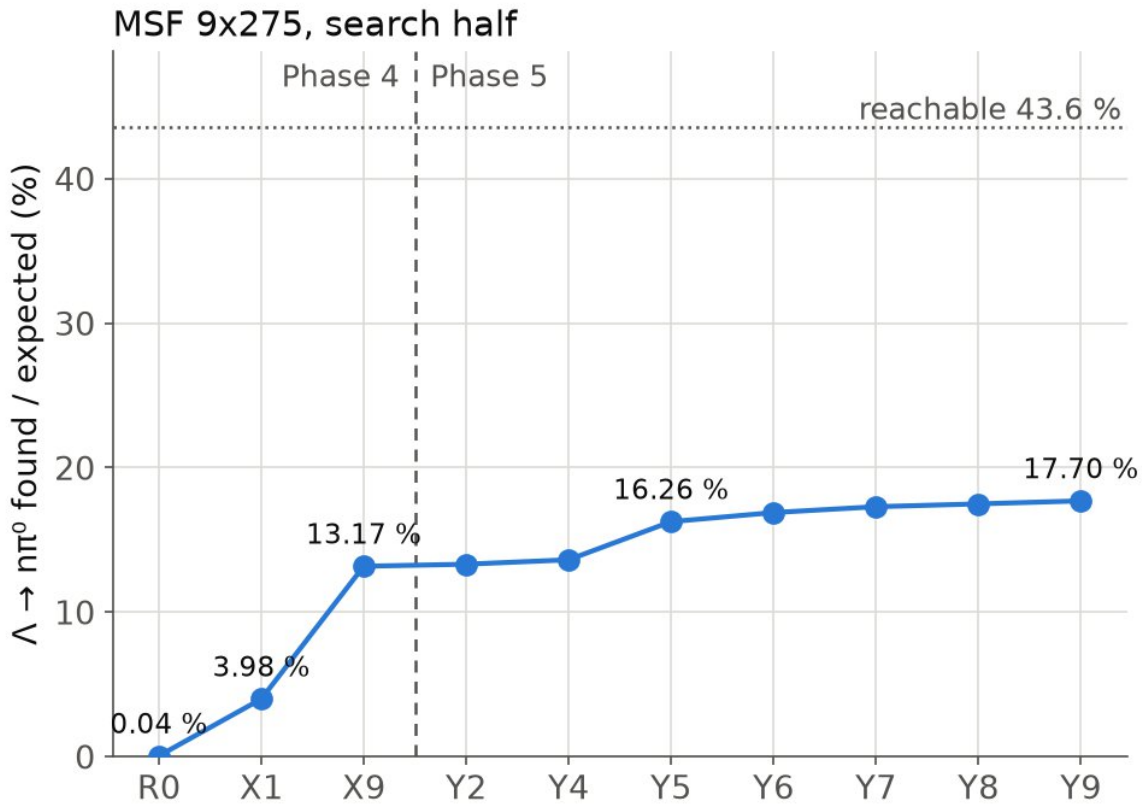
**b 28, c 123, z +7.73; purity z_P +2.48**

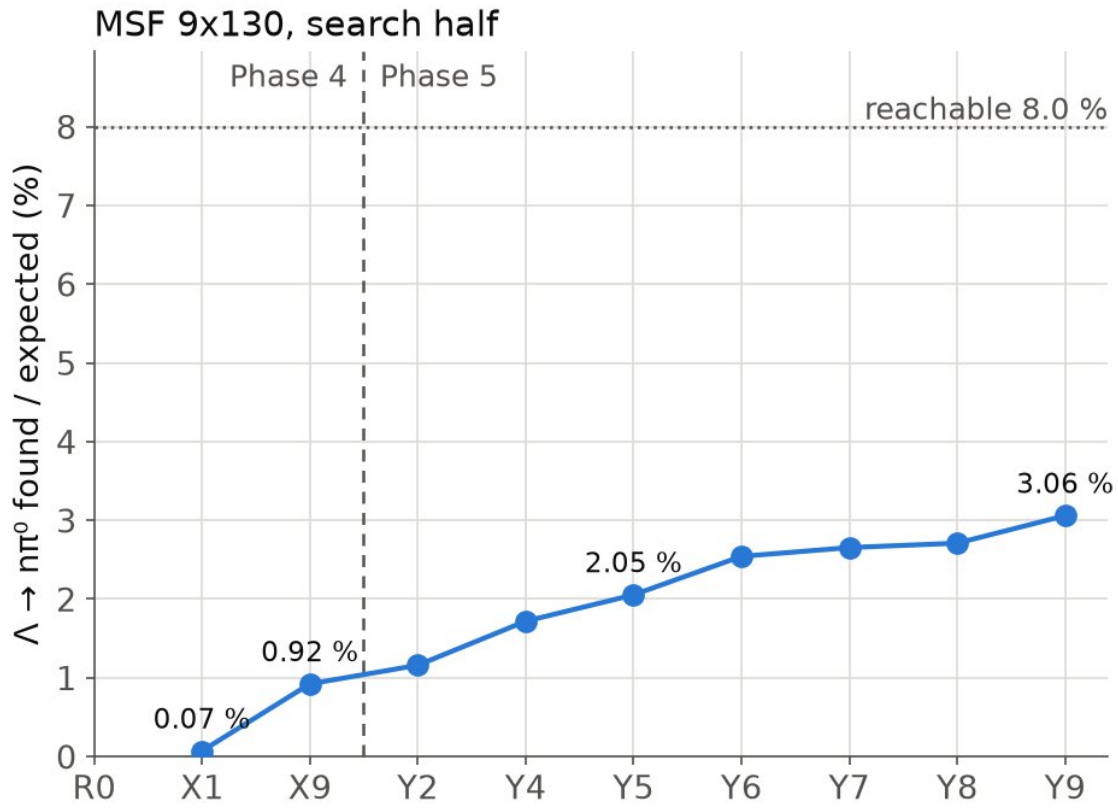
**D4 Λ gun, S4a_tm per thrown Λ → nπ⁰ (reported, never selected on)**

- X9 (end of Phase 4): 7.44 % mean, 14.10 % at 275 GeV;  Y9: 7.44 % mean, 14.12 % at 275 GeV (ceilings 14.1 % / 27.8 %)

**Y7–Y9 measured again on files nothing was tuned on.**

# Phase 4 and Phase 5 on the MSF score — yield step by step

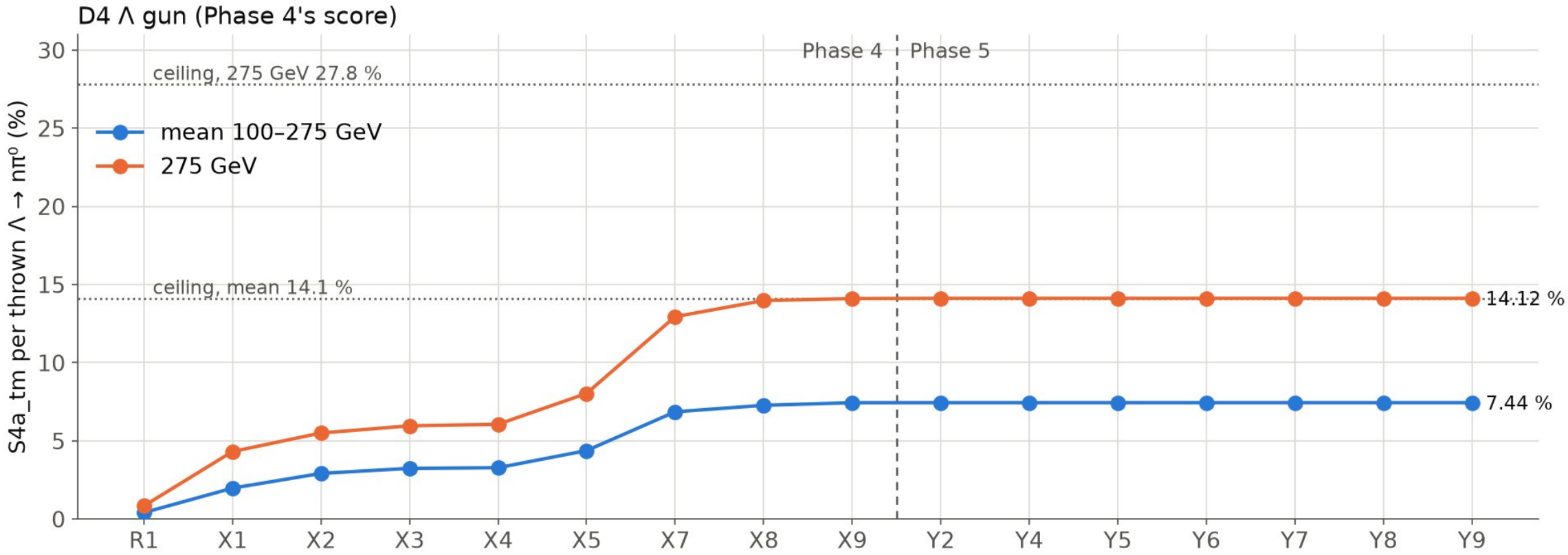



*Λ → nπ⁰ found per expected on the MSF search halves (210). R0 the pin; X1, X9 Phase 4 rows (tuned on the D4 gun); Y2–Y9 the accepted Phase 5 tries (tuned on MSF). R0 was not run at 9x130.*

**Phase 4 took MSF 9x275 from 0.04 % to 13.2 % without seeing MSF; Phase 5 added 4.5 points there and tripled 9x130.**

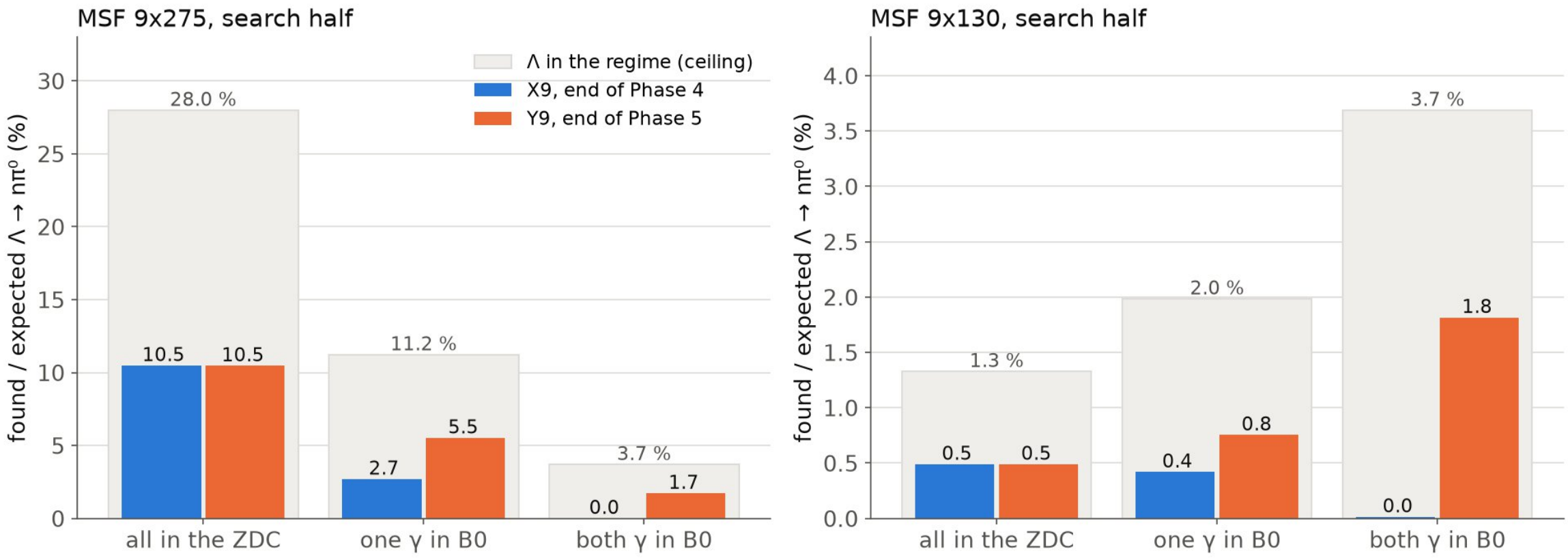
# Phase 4 and Phase 5 on the D4 gun score — S4a_tm step by step




*S4a_tm per thrown Λ → nπ⁰ on the D4 Λ gun (headline.csv): Phase 4 tries R1 → X9, then every accepted Phase 5 try (D4P5_*, CP5_1_Y6_D4, CP5_2_Y9_D4).*

**Phase 5 leaves the gun where Phase 4 left it: its gains are in B0 regimes the gun barely reaches.**

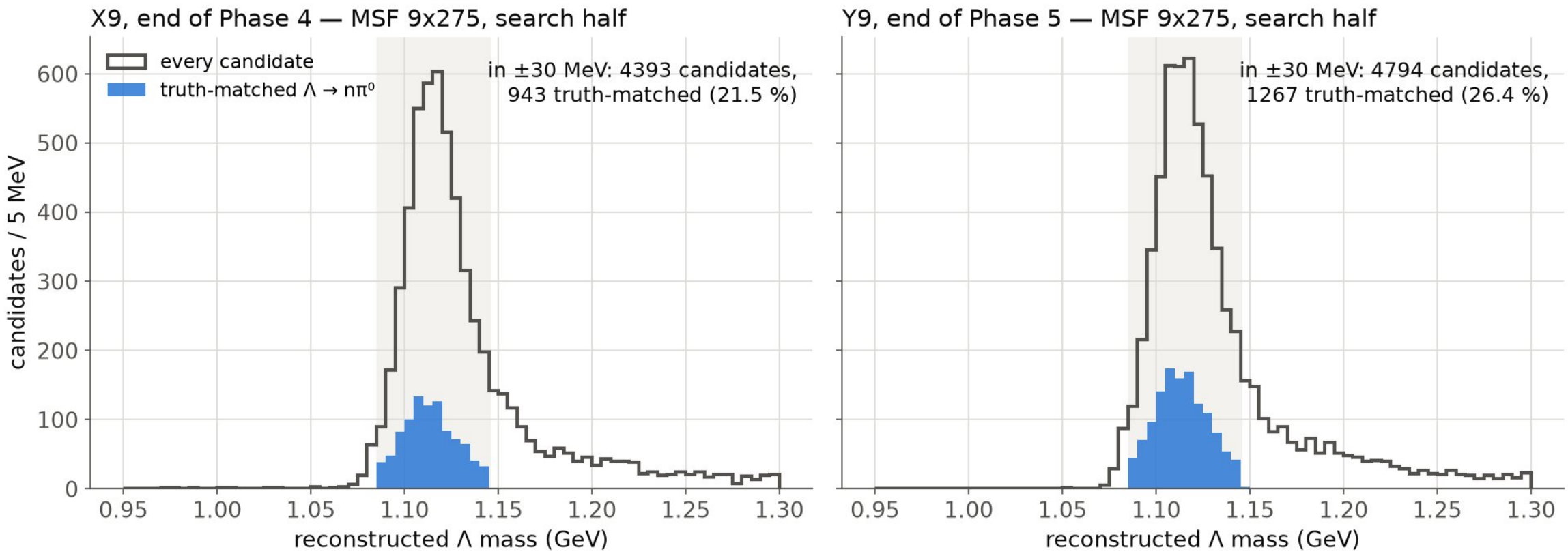
# Phase 5's gain by regime — X9 and Y9 on MSF




*Found per expected Λ → nπ⁰ in each regime (210 groups, search halves); grey: the regime's share of expected, the most any reconstruction could find there.*

**All-ZDC unchanged; one photon in B0 doubled; both photons in B0 from zero to 1.7 % (9x275) and 1.8 % (9x130).**

# Λ mass on MSF 9x275 — every candidate and the truth-matched ones, X9 and Y9




*ReconstructedLambdas mass, at most one candidate per event (F3a), MSF 9x275 search half; shaded: the ±30 MeV scoring window; blue: candidates 210 truth-matches to the true n, γ, γ.*

**The wrong candidates peak at m_Λ too (both pairings rank by Λ mass): purity needs other variables — a phase of its own.**

# Phase 4 and Phase 5 side by side

| | Phase 4 — the Λ bar fight | Phase 5 — Λ → nπ⁰ on MSF events |
|---|---|---|
| data it tuned on | D4 Λ gun (8 momenta × 2000) | MSF 2026-11 search halves (2 × 20 000 events) |
| score | S4a_tm per thrown Λ → nπ⁰ (D4) | N_found on MSF, purity as a floor |
| held out | D4h (fresh seeds) | D6h (MSF files 41–80) |
| tries | 9: 8 accepted, 1 rejected | 9: 7 accepted, 1 inconclusive, 1 rejected |
| D4 S4a_tm, mean / 275 GeV | 0.404 % / 0.832 % → 7.436 % / 14.101 % | X9 → Y9: 7.439 % / 14.117 % (reported, not selected on) |
| MSF 9x275 found / expected (search) | R0 0.04 % → X1 4.0 % → X9 13.2 % | X9 13.2 % → Y9 17.7 % (held out 12.5 % → 16.8 %) |
| MSF 9x130 found / expected (search) | X9 0.92 % | X9 0.92 % → Y9 3.06 % (held out 0.88 % → 2.86 %) |
| purity in the window, 9x275 / 9x130 | X9 21 % / 9 % | Y9 26 % / 29 % |
| what changed | ZDC clustering, photon tag, Λ pairing | vertex lever arm, pairing with B0 photons, B0 splitting and law |

**Next: Phase 6, the crystal-seeded ZDC photon finder (prepared, loop v6)**

- the all-ZDC regime loses a photon for 1125 of 2003 Λ (9x275 half); a scope study decides first (stop below +50 emulated)